

# Calculation Answers

|             |           |           |                                                                                                                                |            |             |             |
|-------------|-----------|-----------|--------------------------------------------------------------------------------------------------------------------------------|------------|-------------|-------------|
|             |           |           | Example values below<br>(other combinations are possible, but the last two columns should be nearly equal for any combination) |            |             |             |
|             | Mass (kg) | Average F | $\Delta v$                                                                                                                     | $\Delta t$ | $m\Delta v$ | $F\Delta t$ |
| Condition 1 | 1.8 kg    | ~0.73 N   | -0.31 m/s                                                                                                                      | 0.81 s     | 0.56        | 0.59        |
| Condition 2 | 2.3 kg    | ~0.93 N   | -0.44 m/s                                                                                                                      | 1.05 s     | 1.02        | 0.98        |
| Condition 3 | 2.3 kg    | ~1.71 N   | -0.44 m/s                                                                                                                      | 0.56 s     | 1.02        | 0.96        |
| Condition 4 | 1.8 kg    | ~1.34 N   | -0.5 m/s                                                                                                                       | 0.7 s      | 0.9         | 0.94        |